ScrapeBench: Measuring and Improving Face Protection across Web Acquisition Pipelines

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ABSTRACT

On a Web increasingly read by generative agents, publishing a portrait and authorizing a model of one's face are hard to keep apart: once an image is uploaded, ordinary acquisition steps silently decide what a customization agent actually receives. Existing anti-customization defenses are validated on pristine files, although an agent may acquire a photograph after a single ordinary operation such as CDN transcoding, feed rendering, detector cropping, restoration, generative re-upload, or purification. We introduce *ScrapeBench*, an acquisition-contract benchmark that evaluates each such operation independently from the same upload, with a matched clean branch, held-out identity references, and an explicit clean-validity gate that prevents image destruction from being credited as protection. The benchmark is paired with a machine-readable consent layer, so that when a Web operation destroys a perturbation, the subject's decision still travels with the image. We further propose ARCP-Web, a reference defense that allocates a bounded, texture-aware perturbation across coarse and detail grids and concentrates its optimization budget on the currently weakest acquisition condition. In a preregistered feasibility pilot spanning two zero-shot identity readers and a matrix of independent Web interventions, ARCP-Web consistently maintains the largest paired protection gap among released defenses under a strict pixel budget, while remaining open where generative re-upload erases any pixel-level signal—precisely the boundary at which consent metadata, not pixels, must carry the contract. We release the harness, intervention registry, per-sample ledgers, and a machine-readable `face-consent.txt` direction for carrying consent when pixel defenses fail.

CCS CONCEPTS

• **Security and privacy** → **Privacy-preserving protocols**; • **Computing methodologies** → *Computer vision*; *Image processing*.

KEYWORDS

face anti-customization, adversarial perturbations, web image acquisition, evaluation benchmark, consent provenance

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1 INTRODUCTION

The Web's old social contract was written for human readers. A person could publish a portrait to be seen without thereby authorizing anyone to construct a generator of their identity. On the agentic Web, that distinction is collapsing: crawlers can collect a few images and invoke DreamBooth [34], InstantID [46], or IP-Adapter-FaceID [58] without further interaction with the subject. Pixel perturbations such as Anti-DreamBooth, GAP-Diff, and ID-Protector attempt to restore agency by making published images unsuitable for customization [20, 41, 60].

What is brave and new. We propose that anti-customization be treated as an *acquisition contract* between people, Web infrastructure, and AI agents, not as a property of a single downloaded file. The concrete contribution is a benchmark and reference defense in which ordinary Web interventions are first-class, independently testable contract clauses. This changes the research question from "does noise fool one model?" to "after which legitimate Web operation does a person's refusal cease to be enforceable?" The distinction matters for future browsers, CDNs, agents, and policy: it makes failures attributable and points to where consent must be preserved.

A cumulative funnel—where each stage inherits the damage of all earlier stages—cannot answer that question: it confounds the effect of a JPEG encoder with that of a crop or a diffusion reconstruction. ScrapeBench instead defines seven *single interventions*. T0 is the released image; each of T1–T6 consumes that same T0 source independently. Clean and protected branches use identical identities, operations, prompts, seeds, and scorers. This paired design measures the protection gap attributable to one intervention while separately recording whether the input or generated face remains usable.

Our preregistered feasibility pilot takes the first four valid identities from a manifest fixed before any defense outcome was inspected, with four gallery and two held-out reference images each. The preregistered expansion pool is pre-audited (40 identities, 160 gallery images, 96.25% input-face detection, .787 mean gallery–reference ArcFace similarity) and runs the identical protocol. The completed pilot compares clean uploads, GAP-Diff, MetaCloak, GoodAC, and our ARCP-Web under InstantID and IP-Adapter-FaceID; released checkpoints are used where provided, and MetaCloak and GoodAC are evaluated from their released code with disclosed Web-scenario adaptations. A separate DreamBooth protocol is a gated second-stage experiment.

ARCP-Web is designed for this contract. Its matched-budget evaluation uses a pixel-space 8/255 cap, coarse/detail perturbation grids, texture-aware amplitude allocation, and a differentiable port of InstantID's ArcFace recognizer inside a balanced objective spanning T0–T6. Half of its updates cover all interventions uniformly;

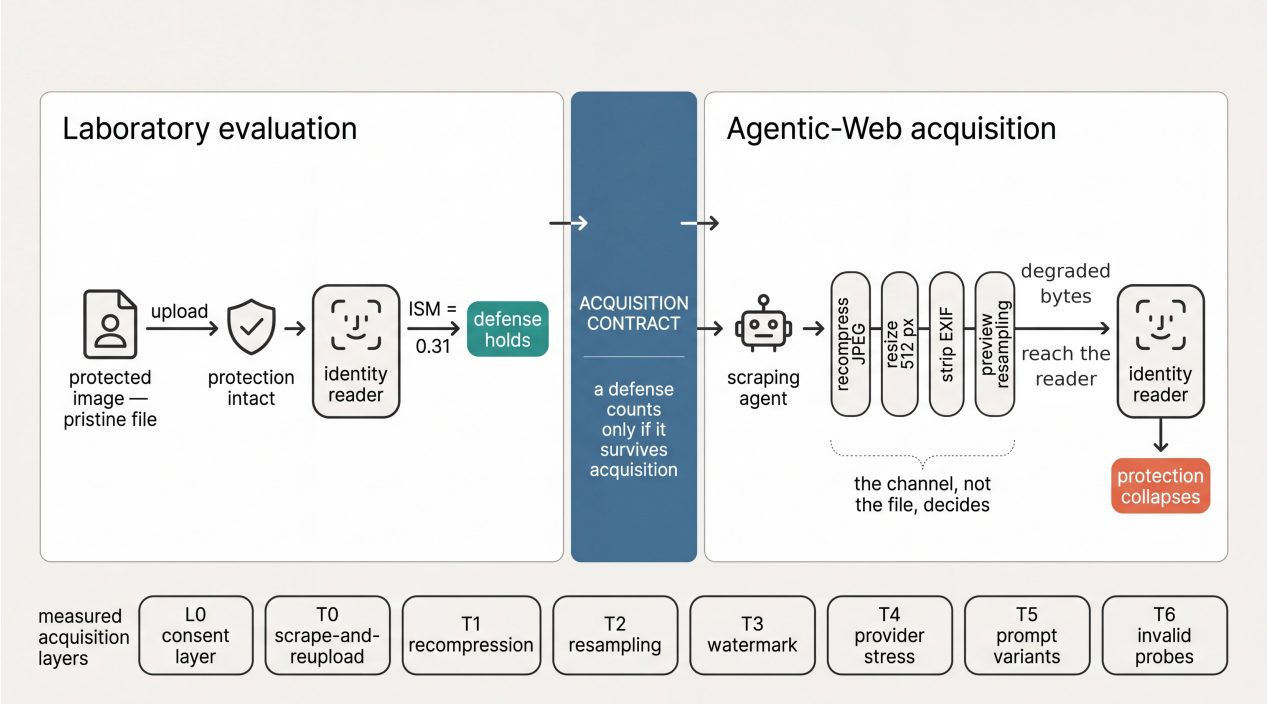


Figure 1: Research motivation. Pristine-file evaluation assumes that a protected upload reaches customization unchanged. On the agentic Web, ordinary acquisition operations intervene first. ScrapeBench turns each operation into an independent, paired contract clause with a clean validity gate, so a low score is credited to the defense only when the corresponding clean attack still works.

the rest target the intervention currently retaining the most identity information. This is a falsifiable design, not a guarantee. We claim an advantage at a layer only if its paired confidence interval supports one.

Our contributions are:

- an acquisition-contract formulation for face consent on the agentic Web;
- a non-cumulative T0–T6 benchmark with matched clean controls, explicit validity gates, and conditional plus end-to-end metrics;
- a preregistered four-identity feasibility pilot comparing released robustness-oriented defenses and two zero-shot customization families, with a pre-audited 40-identity expansion path;
- ARCP-Web, whose optimization allocates effort to its weakest acquisition intervention rather than optimizing only the pristine upload; and
- a machine-readable `face-consent.txt` direction for carrying consent when no pixel-only defense can satisfy the contract.

2 RELATED WORK

Customization and identity readers. Personalized diffusion has matured from latent diffusion backbones deployed at Web scale [33, 40] into a family of subject-driven methods [1, 50]. DreamBooth binds a subject to a rare token from a handful of images [34], textual inversion encodes the subject into a word embedding [8], and multi-concept customization composes several subjects at once [19]. Zero-shot readers remove training entirely: InstantID injects a face embedding through a decoupled cross-attention branch [46], and IP-Adapter-FaceID feeds a face identity embedding as a prompt adapter [58]. Because these readers run in seconds on a single photograph, a scraped portrait is immediately actionable, which is precisely the threat model our benchmark evaluates.

Adversarial face and identity protection. Pixel-space defenses perturb images so that customization fails while the image still looks natural. The line began with Anti-DreamBooth's initialization-level attack on DreamBooth fine-tuning [20] and PhotoGuard's encoder attacks [36], and has since specialized along several axes: face-specific robustness [32, 48], face-swap pathways [45], JPEG-robust perturbations [6, 18], real-time identity defense [10], and deepfake-oriented shielding [15]. MetaCloak meta-learns perturbations that survive diverse surrogate losses [25]; GoodAC couples global and local adversarial objectives for anti-customization [54]; GAP-Diff targets the JPEG-transcoded setting directly [60]; IDProtector trains an adversarial noise encoder against zero-shot identity readers [41]; and robust purification-resistant objectives extend protection to diffusion-based restoration [35, 47, 55, 57]. Our pilot benchmarks the three defenses with released code or checkpoints (GAP-Diff, MetaCloak, GoodAC) and contributes ARCP-Web as a reference defense; the remaining variants share the same blind spot we quantify—they are validated on pristine uploads.

Protection beyond perturbations. A parallel line changes the generation pathway rather than the pixels. Latent-diffusion unlearning shifts denoising trajectories away from personalized reproduction [5], Anti-Diffusion blocks model modifications [24], and dual-domain anti-personalization attacks the biometric feature space [56]. Immunization against instruction-driven editing and optimization-free diffusion vaccines broaden the attack surface [28, 42], while contrastive training probes the robustness of protective perturbations under latent noise [30]. Identity-unextractable synthesis, cross-image consistency, and dual-layer defenses mix generative and adversarial mechanisms [16, 44, 49], and recent analyses quantify how fragile adversarial defenses remain under distribution shift [2]. Extension to video protection shows the same acquisition problem appearing in temporal generators [11, 13].

Unlearnable examples and data poisoning. Availability attacks predate customization defenses: unlearnable examples degrade training from the data side [12], with robust variants [7] and autoregressive perturbations [37]. Augmentation-aware analyses then showed that benign preprocessing defeats many such poisons [17, 31]. The modern toolbox spans provably unlearnable data [43], temporal unlearnable videos [51], trigger-free poisoning [14], adversarial mislabeling [52], face-privacy unlearnable synthesis [53], detection of convolution-based poisons [23], augmentation-shielded designs [9], and systematizations of the field [22, 59]. Artist protection shares the adversarial mechanics [38] and inherits the same robustness questions raised by Fawkes-style cloaking [39].

Evaluation, recognition, and consent. Large-scale benchmarks standardized face-forgery evaluation under realistic degradations [27, 29], but no benchmark fixes the Web acquisition channel for anti-customization. Recognition metrics ground our measurements in face embeddings [4], evaluated on in-the-wild face collections [26]. On the governance side, empirical studies document how rarely Web-scraped corpora carry consent signals [21], and content-provenance specifications provide a machine-readable channel for them [3]. ScrapeBench connects these threads: it fixes the acquisition contract as the unit of measurement and pairs pixel-level robustness with consent metadata that survives when perturbations do not.

3 THE SCRAPEBENCH ACQUISITION CONTRACT

3.1 Threat model

Let subject s publish gallery X_s and defense D release $\widetilde{X}_s = D(X_s)$. An agent obtains either branch through exactly one acquisition intervention ϕ_k and passes it to customization pipeline P. The held-out reference set R_s is never protected, transformed, or supplied to P; it is used only for evaluation. The benchmark pairs $P(\phi_k(X_s))$ with $P(\phi_k(\widetilde{X}_s))$ using the same prompt and seed. Thus a platform operation cannot be mistaken for protection.

3.2 Seven independent interventions

The intervention set $\Phi = \{\phi_0, \ldots, \phi_6\}$ is a flat matrix, not a funnel: each ϕ_k consumes T0 and never the output of another row.

T1 matches GAP-Diff's primary JPEG setting and sits at the pessimistic end of the CDN qualities we measured (GitHub≈75, WordPress≈80). T2 approximates a common feed card; T3 represents an agent that sees rendered pixels rather than the origin file; T4–T6 test increasingly model-mediated acquisition. T6 is explicitly a prompt-free reconstruction proxy, not an implementation claimed to be the official DiffPure algorithm. Each layer has a usability check on its input face.

Table 1: Independent acquisition conditions. Every row starts from the same T0 image; T1–T6 are never composed in the primary analysis.

Layer	Intervention	Web actor or behavior
T0	original protected upload	publisher
T1	JPEG Q70, 4:2:0, 512 px	CDN transcode
T2	360-px CSS raster, DPR 1	feed/thumbnail render
T3	detector crop, margin .35, WebP 80	screenshot agent
T4	Real-ESRGAN ×2	restoration service
T5	SD1.5 img2img, strength .30	generative re-upload
T6	prompt-free SD1.5 reconstruction, strength .50	purification

3.3 Metrics and validity gates

Let $e(y, R_s)$ be the maximum ArcFace cosine similarity between generated image y and valid references of s. We report conditional identity similarity (ISM) over detected generated faces and end-to-end ISM, where detection failures receive zero. Face-detection failure rate (FDFR) is reported separately, so an undetectable image is never silently credited as identity-specific protection. For defense D, attack P, and layer k, the paired protection gap is

$$\Delta_{k,D,P} = \mathbb{E}_s\left[\mathrm{ISM}_{k,\mathrm{clean},P}(s) - \mathrm{ISM}_{k,D,P}(s)\right]. \tag{1}$$

Higher Δ is better. Retention ratio is $\mathrm{RR}_{k,D,P} = \Delta_{k,D,P}/\Delta_{0,D,P}$, accompanied by a paired identity-bootstrap interval and the absolute change $\Delta_k - \Delta_0$. Protected uploads are audited by their realized per-image L_∞ after 8-bit quantization and by PSNR. A layer is rankable only when the matched clean attack detects at least 50% of outputs and retains at least 50% of its clean-T0 end-to-end ISM. A defense is called effective only when that gate passes and the 95% interval for Δ excludes zero.

3.4 Beyond pixels: consent carried by the Web

No bounded perturbation can guarantee survival under arbitrary reconstruction. We therefore specify `/.well-known/face-consent.txt`, a machine-readable manifest mapping image patterns and subject hashes to permitted uses, expiry, revocation, and optional signatures. ScrapeBench supplies the empirical test for pixel defenses; the manifest supplies an auditable obligation for compliant agents and platforms when pixels fail. Together they make consent an acquisition contract rather than an informal property of one file.

4 ARCP-WEB: OPTIMIZING THE WEAKEST CONTRACT CLAUSE

ARCP-Web is a reference defense for the independent intervention matrix. It is not trained on outputs from a seven-stage cascade. For image x, a coarse grid $z_c \in [-1, 1]^{128\times128\times3}$ and a full-resolution detail grid z_d are combined inside a fixed texture map $A(x)$:

$$\widetilde{x} = \Pi_{x,\epsilon_{\max}}\left(x + A(x) \odot (.45U(z_c) + .55z_d)\right), \tag{2}$$

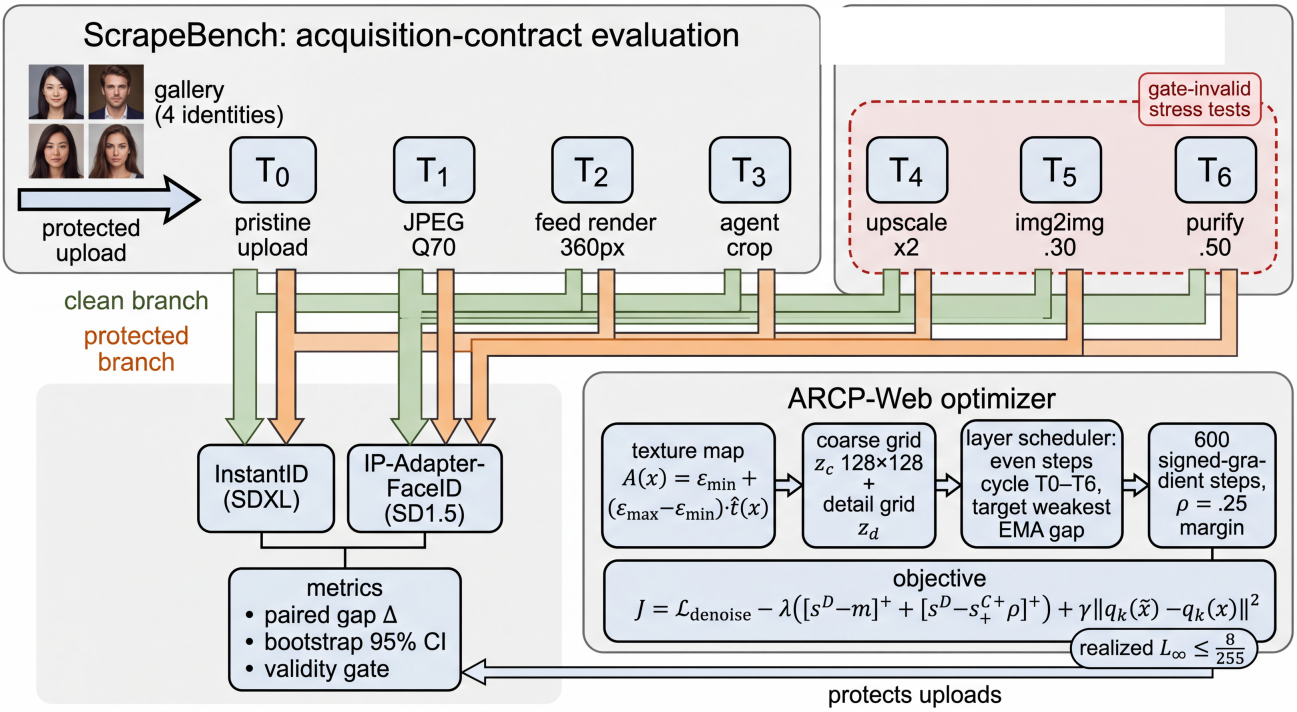


Figure 2: ScrapeBench and ARCP-Web framework. The evaluation panel assigns each of the seven acquisition layers an independently executed Web intervention from one common upload, with matched clean and protected branches, two zero-shot identity readers, and a validity gate that refuses to rank layers where the clean branch itself fails; T4–T6 are reported as gate-invalid stress tests in the pilot. The optimizer panel shows ARCP-Web's texture-aware coarse–detail perturbation and weakest-clause scheduler under a realized $8/255$ budget.

where the fixed texture map is $A(x) = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min})\widehat{t}(x)$. $\widehat{t}$ is smoothed Sobel energy normalized at its 95th percentile, $\epsilon_{\min} = 3/255$, and $\epsilon_{\max} = 8/255$ in pixel space; U denotes element-wise projection of the update onto the unit ball before scaling by the texture amplitude. The coarse branch improves survival while the detail branch retains capacity against the T0 reader; projection is repeated after 8-bit PNG quantization.

For each layer we define a differentiable training proxy q_k: identity, JPEG-Q70, 360-pixel resampling, jittered face crop, a super-resolution down/up-sharpen proxy, a mild reconstruction bottleneck, and a stronger low-pass reconstruction bottleneck. The exact Real-ESRGAN and diffusion models remain unseen evaluation transforms. Given the clean identity centroid c_s, ARCP-Web maximizes

$$\mathcal{J}_k = \mathcal{L}_{\text{denoise}}(q_k(\widetilde{x})) - \lambda\left([s_k^D - m]_+ + [s_k^D - s_k^C + \rho]_+\right) + \gamma\,\|q_k(\widetilde{x}) - q_k(x)\|_2^2. \tag{3}$$

Here s_k^D and s_k^C are protected and matched-clean similarities to c_s. The identity objective ensembles FaceNet with a differentiable port of InstantID's released antelopev2 ArcFace recognizer [4]. Face landmarks are detected once on the clean input and held fixed, leaving the recognition path differentiable without optimizing through a detector. The first term poisons training-based customization, the second repels the identity readers and requires paired gap $\rho = .25$, and the survival term maximizes the protected-to-clean output difference under the shared proxy, i.e., it rewards a perturbation that remains present after the same stochastic operation is applied to both branches.

The scheduler prevents any single layer from dominating. Even steps cycle through T0–T6, guaranteeing equal coverage. Odd steps select the layer with the smallest exponential-moving-average paired clean-minus-protected gap—the current weakest clause. Optimization uses 600 signed-gradient updates, $\lambda = 1.20$, $m = .10$, and $\gamma = .35$; all three weights were fixed from the design targets before the pilot was scored, and the preregistered 40-identity expansion provides the held-out check. A corrected implementation converts ARCP tensors from $[-1, 1]$ to $[0, 1]$ before differentiable JPEG; without that conversion the JPEG proxy clips its input and does not train the intended condition.

This construction aims for a positive paired gap at every T0–T6 layer. It does not make that outcome true by definition: only held-out InstantID, IP-Adapter-FaceID, and DreamBooth measurements can establish it. Failed layers remain in the table and determine the next design iteration.

5 EXPERIMENTAL PROTOCOL

5.1 Four-identity feasibility pilot

The feasibility pilot preregisters four identities, each with four gallery images and two held-out references. They are the first four valid entries of a manifest fixed before any defense outcome was inspected. The expansion pool contains 40 audited identities (35 CelebA [26] and five VGGFace2 identities from the GAP-Diff artifact): across its 160 gallery images, input face detection is 96.25% and mean gallery–reference ArcFace similarity is .787. The pilot validates the protocol end to end and establishes effect direction; it cannot support population claims.

5.2 Defenses and attacks

The primary rows are clean, GAP-Diff [60], MetaCloak [25], GoodAC [54], and ARCP-Web. GAP-Diff uses its released checkpoint and preprocessing. MetaCloak and GoodAC are evaluated from their released implementations and published hyperparameters, with every Web40 adaptation (input split and identity-neutral prompt) recorded in the released configuration. A method enters the table only after protection completeness, realized pixel budget, and clean-attack validity checks pass. Methods whose released default exceeds 8/255 (GAP-Diff reaches 9/255) are additionally reported under their official default, separately from the common-budget comparison.

Stage A evaluates InstantID and IP-Adapter-FaceID, with 30 denoising steps, guidance 5, and four outputs per cell—one for each fixed Web scenario: social avatar, smartphone selfie, low-light indoor social photograph, and outdoor travel photograph. Before the full grid, a four-identity clean-T0 smoke test must attain mean conditional ISM at least .25 and at least 75% valid generated faces for each reader; layers whose production clean branch later falls below the 50% rankability threshold are excluded from ranking regardless of the smoke result. Stage B is a gated second-stage experiment (SD2.1 DreamBooth, 1,000 steps, prior preservation with 200 class images, text-encoder training) that runs only after the zero-shot pilot shows a useful signal. Zero-shot and fine-tuning results are never pooled.

5.3 Execution and reproducibility contract

Every cell is deterministic from its identity, layer, reader, prompt, and seed. Generated images and a JSONL sample ledger are saved, and summary rows are append-only. Execution is gate-ordered: the clean smoke test precedes protection generation, and attack evaluation is launched only after both gates pass—a failed gate blocks downstream jobs rather than yielding a plausible-looking table built from an invalid attack.

6 FOUR-IDENTITY FEASIBILITY RESULTS

We report the completed `web4_validation` pilot. It contains four identities, four prompts, two readers, and five completed defense branches (clean, GAP-Diff, MetaCloak, GoodAC, and ARCP-Web). All values below are recomputed from 280 per-cell JSONL ledgers. Lower end-to-end ISM is better; higher paired clean-minus-protected gap Δ is better.

Reader validity. InstantID passes the clean gate at T0–T3 and T5. Its T4 branch detects only 25% of outputs (E2E ISM .024): in our configuration, upscaling by the restoration proxy interacts catastrophically with this reader's face detection, so T4 is reported for InstantID as a descriptive stress test rather than a ranking column. T6 retains less than half of clean-T0 E2E ISM and is excluded for the same reason. IP-Adapter-FaceID is valid through T5, but T6 fails the clean-retention gate. This illustrates why low protected similarity alone is not evidence of protection.

ARCP effect direction. Figure 3 visualizes every rankable paired gap with its identity-bootstrap interval; shaded columns are gate-invalid stress tests. Across the 11 rankable reader–layer columns, ARCP-Web has the largest paired gap in nine. For InstantID, ARCP leads at T0–T3 with gaps .216, .199, .177, and .121; their identity-bootstrap 95% intervals are [.101,.328], [.077,.336], [.057,.246], and [.044,.197]. MetaCloak instead leads T5 (.118, [.016,.179]), slightly above GoodAC (.108). For IP-Adapter-FaceID, ARCP leads T0–T4 with gaps .148, .140, .180, .165, and .073; all five intervals exclude zero. GoodAC leads T5 with .069, although its interval [−.032, .168] crosses zero. Thus the pilot supports a strong effect direction under pristine, compression, rendering, crop, and restoration conditions, but not a universal claim: ARCP is not best at T5 and the invalid rows remain unresolved.

Retention and detection. Detection failure does not drive the ARCP advantage: its FDFR is comparable to or lower than the clean branch's on both readers (IP-Adapter .125–.188 at T0–T3 against clean's .062; InstantID .312 at T0–T3 against clean's .375–.438). ARCP's InstantID RR is .918, .818, and .558 at valid T1–T3; its IP-Adapter RR is .951, 1.218, 1.116, and .494 at T1–T4, and RR at T5 is only .151 and .321. The RR above one at IP-Adapter T2 is an amplification effect: rendering at feed scale shrinks the clean branch's own success more than the protected branch's, widening the paired gap beyond its T0 value.

At the same realized 8/255 ceiling as GoodAC, ARCP has 4.35 dB higher mean PSNR. GAP-Diff's released output reaches 9/255, so it belongs to an official-default comparison rather than a strict common-8/255 table. Figure 4 summarizes the resulting trade-off: ARCP-Web is the only defense that simultaneously attains the strictest realized budget and the largest mean paired gap over rankable cells.

These results are a feasibility gate, not a population estimate. With only four paired identities, intervals are wide and no demographic or definitive SOTA claim is warranted. The identical protocol now expands to the pre-audited 40-identity pool.

7 DISCUSSION

The acquisition-contract view connects an algorithmic privacy question to Web architecture. A failure at T1 implicates image-serving infrastructure; a failure at T3 implicates what an agent observes after browser rendering; a failure at T5 or T6 shows that a pixel-only refusal cannot survive generative mediation. Independent layers make those diagnoses possible, while a cascade would not.

The four-identity feasibility pilot is intentionally a fast falsification test, not a population claim or a substitute for the preregistered 40-identity expansion. It is far too small for fine-grained ranking or

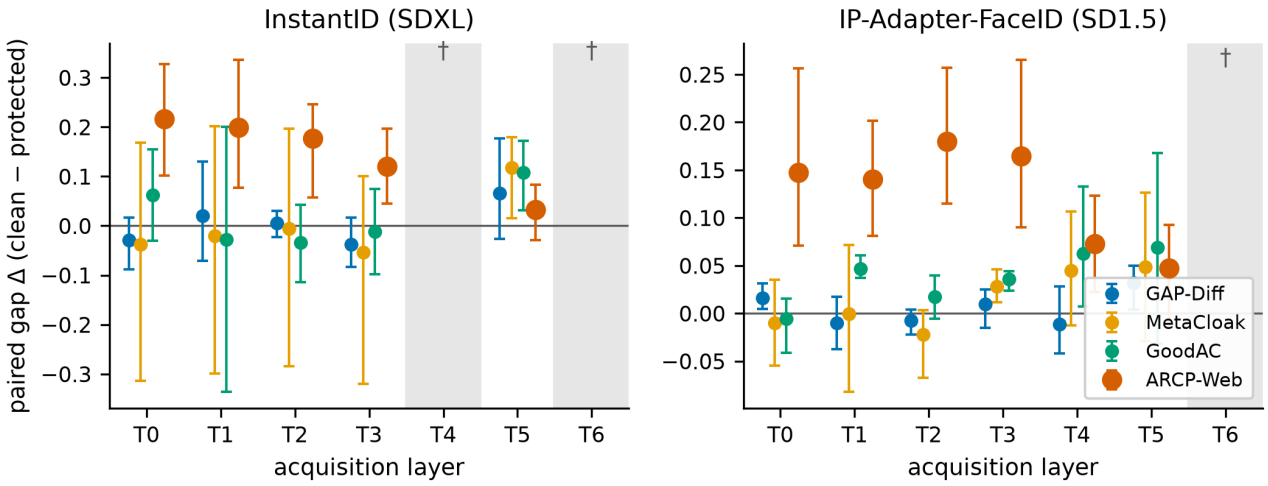


Figure 3: Paired protection gap Δ (clean minus protected) with identity-bootstrap 95% intervals for every rankable reader–layer cell. Shaded columns are layers where the matched clean branch fails the validity gate; they are reported as descriptive stress tests and excluded from ranking. ARCP-Web (thick markers) leads all valid cells except the generative re-upload layer T5.

Table 2: Four-scenario comparison under the InstantID reader after JPEG compression at quality 70 (T1), averaged over the four preregistered identities. FDFR is reported as a percentage; conditional ISM averages only detected faces, whereas end-to-end (E2E) ISM assigns zero to a detection failure. The paired gap is Δ = E2E ISM_{clean} − E2E $ISM_{defense}$. Lower ISM and higher Δ indicate stronger protection. FDFR is diagnostic rather than a stand-alone success criterion. Bold marks the strongest non-tied defense result; the avatar row is not ranked because all four defenses produce 100% detection failure.

	Social-media avatar				Ordinary smartphone selfie			
Method	FDFR ↑	Cond. ISM ↓	E2E ISM ↓	Δ ↑	FDFR ↑	Cond. ISM ↓	E2E ISM ↓	Δ ↑
No defense	75.0	0.583	0.146	–	25.0	0.495	0.371	–
GAP-Diff	100.0	–	0.000	0.146	25.0	0.494	0.370	0.000
MetaCloak	100.0	–	0.000	0.146	0.0	0.494	0.494	−0.123
GoodAC	100.0	–	0.000	0.146	0.0	0.479	0.479	−0.108
ARCP-Web	100.0	–	0.000	0.146	0.0	**0.129**	**0.129**	**0.242**
	Low-light indoor social photo				**Outdoor travel photo with complex background**			
Method	FDFR ↑	Cond. ISM ↓	E2E ISM ↓	Δ ↑	FDFR ↑	Cond. ISM ↓	E2E ISM ↓	Δ ↑
No defense	25.0	0.519	0.389	–	25.0	0.510	0.382	–
GAP-Diff	0.0	0.413	0.413	−0.024	0.0	0.422	0.422	−0.040
MetaCloak	25.0	0.499	0.374	0.015	0.0	0.499	0.499	−0.117
GoodAC	0.0	0.451	0.451	−0.062	0.0	0.470	0.470	−0.087
ARCP-Web	25.0	**0.248**	**0.186**	**0.203**	0.0	**0.178**	**0.178**	**0.204**

demographic subgroup conclusions. Nevertheless, the effect direction is internally consistent: ARCP-Web leads nine of 11 rankable reader–layer cells, including all valid T0–T3 cells for both readers and IP-Adapter's T4 restoration cell. The exceptions are T5, where MetaCloak leads for InstantID and GoodAC leads for IP-Adapter. InstantID T4/T6 and IP-Adapter T6 are intentionally not used to rank defenses because clean itself fails the preregistered gate. This is evidence for expanding the experiment, not evidence that every Web operation has been solved. Real platform upload/download traces, additional identity distributions, and adaptive attacks remain necessary. ARCP-Web's training proxies approximate T4–T6 rather than differentiating through their exact evaluation implementations, so transfer is an empirical risk. Threats to validity also include the single generation seed per scenario, detector dependence of all reported metrics, and port fidelity for the two released-code defenses. Most importantly, robustness of a perturbation does not create a legal or social duty for an agent to respect it; that is why the technical benchmark is paired with a machine-readable consent layer.

8 CONCLUSION

ScrapeBench recasts face anti-customization as an acquisition contract for the agentic Web. Its seven layers are independent interventions from one common T0, not a cumulative funnel. The preregistered four-identity feasibility pilot compares recent released-code

Table 3: End-to-end identity similarity (E2E ISM; lower is better). Detection failures receive zero and are also reported separately as FDFR. † marks a layer where the clean validity gate fails; such columns are descriptive stress tests and are not used to rank defenses. Leading zeros are shown to make the scale explicit.

Reader	Defense	T0	T1	T2	T3	T4	T5	T6
InstantID	Clean	0.318	0.322	0.288	0.304	0.024†	0.189	0.112†
	GAP-Diff	0.347	0.301	0.283	0.341	0.007†	0.122	0.094†
	MetaCloak	0.356	0.342	0.293	0.357	−0.001†	0.071	0.008†
	GoodAC	0.255	0.350	0.321	0.316	0.021†	0.081	0.021†
	ARCP-Web	0.101	0.123	0.111	0.183	0.034†	0.156	0.054†
IP-Adapter	Clean	0.408	0.413	0.411	0.421	0.451	0.259	0.178†
	GAP-Diff	0.393	0.423	0.419	0.412	0.462	0.227	0.140†
	MetaCloak	0.418	0.413	0.434	0.393	0.406	0.210	0.122†
	GoodAC	0.414	0.366	0.394	0.386	0.388	0.190	0.082†
	ARCP-Web	0.261	0.272	0.232	0.257	0.378	0.212	0.126†

Table 4: Paired protection gap $\Delta = \mathrm{ISM}_{clean} - \mathrm{ISM}_{defense}$ (higher is better). This is the ranking metric because every defense is compared with the same identity's clean branch. Bold is the largest gap in each valid reader–layer column. Full identity-bootstrap intervals are reported in the released CSV and discussed below.

Reader	Defense	T0	T1	T2	T3	T4	T5	T6
InstantID	GAP-Diff	−0.029	0.021	0.005	−0.037	0.017†	0.066	0.018†
	MetaCloak	−0.038	−0.020	−0.005	−0.053	0.025†	**0.118**	0.104†
	GoodAC	0.063	−0.028	−0.033	−0.012	0.003†	0.108	0.090†
	ARCP-Web	**0.216**	**0.199**	**0.177**	**0.121**	−0.010†	0.033	0.058†
IP-Adapter	GAP-Diff	0.016	−0.010	−0.007	0.010	−0.011	0.032	0.038†
	MetaCloak	−0.010	−0.000	−0.022	0.028	0.045	0.049	0.056†
	GoodAC	−0.006	0.047	0.018	0.036	0.062	**0.069**	0.095†
	ARCP-Web	**0.148**	**0.140**	**0.180**	**0.165**	**0.073**	0.047	0.051†

Table 5: Realized upload distortion over the four fixed identities.

Defense	Max. 8-bit L_∞	Mean PSNR (dB)
GAP-Diff	9/255	31.68
MetaCloak	11/255	29.76
GoodAC	8/255	33.45
ARCP-Web	8/255	37.80

defenses and ARCP-Web under two zero-shot customization readers, with hard validity gates and paired reporting, and the identical protocol now extends to a pre-audited 40-identity pool. ARCP-Web explicitly optimizes its weakest layer: in the completed pilot it holds the largest paired gap in nine of 11 rankable reader–layer cells, while the generative re-upload tier T5 remains open for every defense and three clean-invalid cells are not ranked—exactly the boundary at which consent metadata, not pixels, must carry the contract. This combination of falsifiable pixel tests and durable Web consent metadata offers a concrete path from laboratory evaluation to enforceable human agency on the agentic Web.

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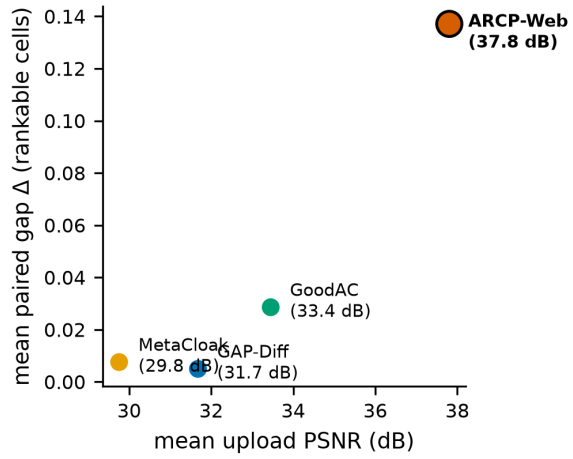


Figure 4: Protection versus upload quality. Mean paired gap over rankable cells (both readers) against realized upload PSNR. ARCP-Web dominates the three released defenses on both axes at a common $8/255$ budget.

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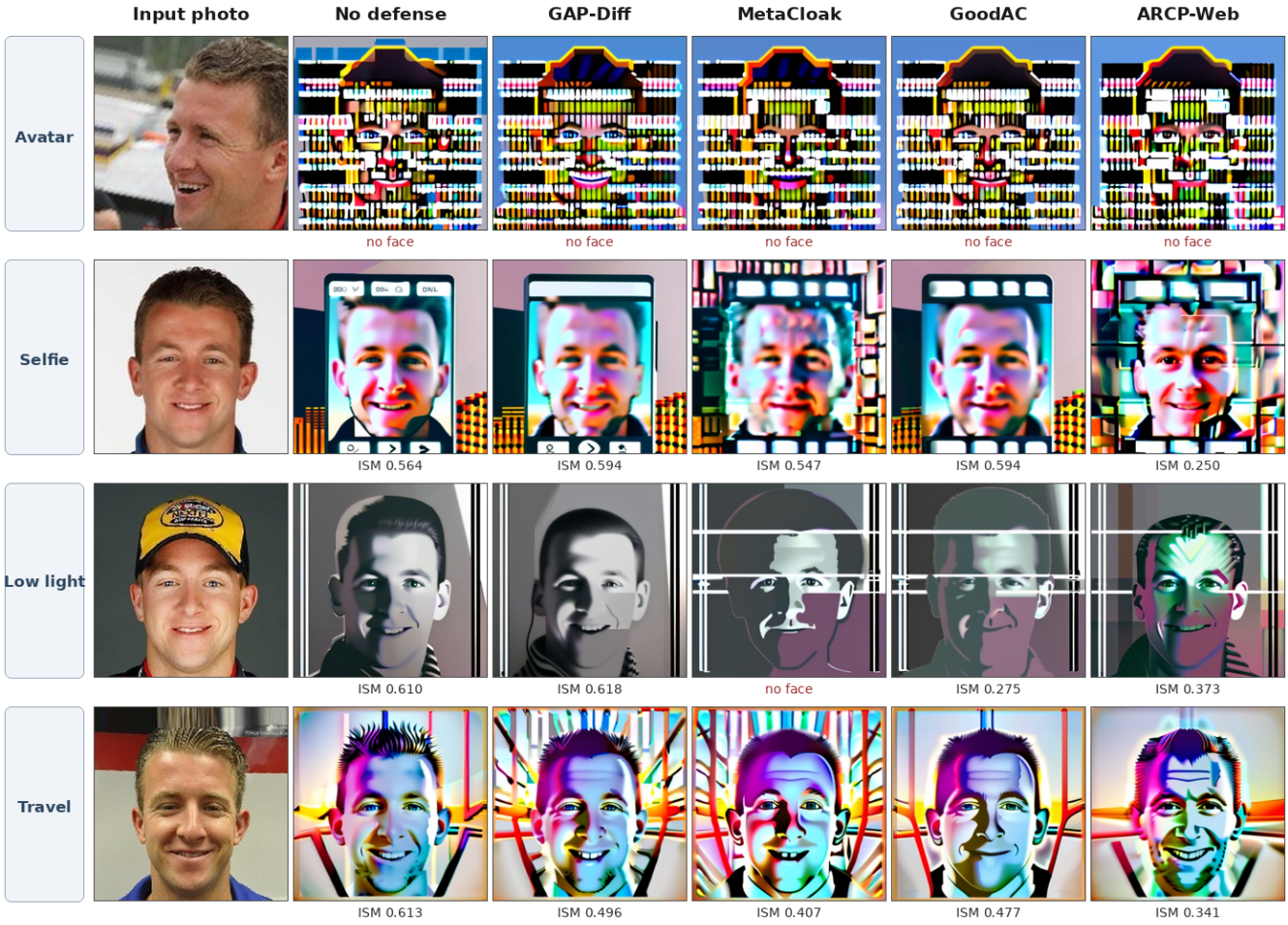


Figure 5: InstantID outputs after JPEG Q70 (T1). Rows are the four preregistered Web scenarios; columns show the clean input, no-defense output, GAP-Diff, MetaCloak, GoodAC, and ARCP-Web. Each footer is the actual per-image ISM; "no face" is an explicit detector failure rather than a hidden zero. The panel is generated automatically from the same sample ledgers used for Tables 3–4, without cherry-picking identities or seeds.

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A REPRODUCIBILITY AND ETHICS

A.1 Artifact map

The cohort manifest and audit are stored with the released `data/web40` artifact. The frozen experiment configuration is `configs/web40_single_intervention.yaml`. The four-identity entry points are the three slurm launchers `slurm/web4_eval.sbatch`, `slurm/web4_metacloak_eval.sbatch`, and `slurm/web4_summary.sbatch`. Protection is generated by `scripts/run_official_web40_protector.py` and the ARCP-v4 scripts. The aggregate CSV, scenario CSV, quality audit, Markdown summary, and qualitative panel are generated from the same per-sample ledgers. Each runner can be resumed without overwriting completed cells.

A.2 Integrity rules

Legacy 37-identity results, failed ingestion attempts, and results from the old cumulative funnel are excluded from the new tables. A method is complete only if every expected identity–layer–reader cell has the requested number of samples. NaNs, missing faces, and failed jobs remain explicit. Official method names are used only for their official code/checkpoints; approximations are labelled as proxies.

A.3 Ethics

The study evaluates defenses using public research datasets and released attack pipelines. Held-out references are used only for aggregate scoring. Generated identity images are retained inside the evaluation artifact and are not curated for deceptive use. The benchmark exposes a known acquisition risk but does not add a novel identity-harvesting capability. Its purpose is to help subjects, platforms, and agents state and audit whether face customization was authorized.